

- synthetic data dictionary · end-to-end operational flow

v2.0 · network graphs + shell detection

⚡ **reconciliation flow** physical vs commercial · unbilled leakage · outbound stipends · **shared directors, fleets & sleeper-cell patterns** injected to expose fraud rings.

I. Master Data · Ground Truth

products.csv

MASTER

COLUMN	DESCRIPTION
product_id	Unique code (PMS, AGO, DPK)
product_name	Official descriptive name
unit_price_kes	Illustrative unit price
is_active	Status flag

depots.csv

FACILITY

COLUMN	DESCRIPTION
depot_id	Natural key (e.g. Nairobi)
depot_name	Formal depot name
location	Geographic mapping
capacity_litres	Storage capacity
is_active	Operational status

omcs.csv

CLIENT REGISTRY

COLUMN	DESCRIPTION
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tariffs.csv

PRICING ENGINE

COLUMN	DESCRIPTION
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omc_id	Primary key
customer_name	Commercial title
kra_pin	Tax ID
payment_terms_days	Contractual window
credit_limit_kes	Financial ceiling
risk_rating	Low/Medium/High
risk_profile	Good, Small, Medium, High
director_name NEW	Shared-director fraud detection
registered_address NEW	Shell-company location
incorporation_date NEW	Sleeper-cell flag
primary_bank_account NEW	IBAN for reconciliation
is_active	Authorized for loading

tariff_id	Version control
product	Fuel grade
depot	KPC facility
pipeline_tariff_per_m3_km	Transport cost
storage_tariff_per_m3_day	Daily holding cost
effective_start	Activation date
effective_end	Expiration

II. Physical vs Commercial Core

depot_loading_logs.csv METER TRUTH	
COLUMN	DESCRIPTION
loading_id	Digital footprint
gantry_bay_id	Hardware location
meter_id	Flow meter

dispatches.csv COMMERCIAL INTENT	
COLUMN	DESCRIPTION
dispatch_id	Official waybill
loading_id	Bridge to meter
date / year / month	Temporal dimensions

dispatch_id	Link to waybill (null = leakage)
omc_id	Client receiving product
depot	Gantry site
product	Fuel classification
physical_loaded_liters	Raw meter volume
temperature_c	Thermal state
density_kg_m3	Measured density
loading_timestamp	Completion time

omc_id / customer_name	Responsible party
product / depot	Fuel & origin
volume_liters	Intended for billing
distance_km	Journey length
truck_reg_number NEW	Shared-fleet detection
driver_id NEW	Driver identifier
transport_tariff_kes / storage_tariff_kes	Computed fees
value_kes	Total commercial worth
dispatched_by NEW	KPC clerk

III. Financial Flow · Uncollected Revenue

invoices.csv BILLING RECORD	
COLUMN	DESCRIPTION
invoice_id	Financial demand note
dispatch_id	Commercial event
omc_id / customer_name	Debtor
product	Product line
date	Issuance date
value_kes	Total expected

payments.csv CASH REALITY	
COLUMN	DESCRIPTION
payment_id	Receipt number
invoice_id	Bill being cleared
omc_id / customer_name	Source of payment
remitting_bank_account NEW	Origin account
date	Cash recognition

prepared_by **NEW** Finance staff

value_kes Amount received

installment_no Partial settlement

IV. Outbound · Inuka Foundation Stage 2

officers.csv

PROGRAM MASTER

COLUMN	DESCRIPTION
officer_id	Primary key
officer_name	Formal name
program_site	Region/site
risk_profile	Leakage behavior
phone / is_active	Contact / status

beneficiaries.csv

PARTICIPANT REGISTRY

COLUMN	DESCRIPTION
beneficiary_id	Unique ID
beneficiary_name	Full name
officer_id	Managing officer
pillar_id	Scholarship/Plus/Vocational/Tech
guardian_name	Registered guardian
registered_address	Location
mobile_money_number NEW	Shared-number fraud ring
enrollment_date / is_active	Entry / eligibility

beneficiary_consents.csv

KDPA

COLUMN	DESCRIPTION
consent_id	Tracking footprint
beneficiary_id	Participant link

attendance.csv

ELIGIBILITY

COLUMN	DESCRIPTION
attendance_id	Primary tracking
beneficiary_id / officer_id	Participant & verifier

consent_type	Photography, SMS, etc.
status	active / withdrawn / expired
captured_at / captured_by	Log & officer
status_updated_at	Change timestamp

pillar_id / period	Program & month
attendance_date	On-site date
verified_at	Officer finalization
eligible_amount_kes	Calculated value
status	Verified

stipend_authorizations.csv SIGN-OFF	
COLUMN	DESCRIPTION
authorization_id	Approval identifier
attendance_id	Physical attendance proof
beneficiary_id / pillar_id	Recipient & program
period / date	Month & formalization
amount_authorized	Approved stipend
authorized_by	Officer ID

disbursements.csv CASH OUTFLOW	
COLUMN	DESCRIPTION
disbursement_id	Outbound receipt
authorization_id	Bridge (NULL = ghost payment)
beneficiary_id / pillar_id	Recipient / funding source
period / date	Month & clearing date
amount_paid	Real cash transferred
channel / disbursing_account	mobile_money / bank
processed_by	HQ payments staff

V. Operational Governance

depot_daily_inventory.csv MACRO RECONCILIATION	
COLUMN	DESCRIPTION

quarantine_audit_log QUALITY SHIELD	
COLUMN	DESCRIPTION

record_id	Audit reference
date / depot / product	Window, terminal, fuel
opening_stock_liters	Start volume
received_pipeline_liters	Fresh volume
dispatched_liters	Exited via gantry
book_closing_stock_liters	Paper expectation
physical_dip_closing_stock_liters	Actual tank dip
variance_liters	Critical delta

quarantine_id	Failed row marker
dataset_name	Upstream table
failed_gate	ETL rule (Duplicate, Null)
reason	Data quality failure
quarantined_at	Interception timestamp
raw_record_json	Corrupted data preserved

-  **KPC colours** · black #1a1a1a · white · red #c8102e
-  **Network graph attributes** shared directors, fleets, temporal sleeper-cell patterns
-  **Stage 2 Inuka** mapped entity-for-entity to inbound processes